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all the sectors I previously worked in, including telecommunications, space, military defence, and test and measurement, are now our customers. This insight gives me a strong appreciation for what happens in their world and why they require these parts. That is why I am excited about our role at Mini-Circuits. We develop the foundational components necessary for numerous applications, including systems where I was once the customer.

**From the customer side, what requirements do you think component vendors must internalise—especially for space and similar systems?**

I would say, the need for better performance, smaller form factors, and high sensitivity to direct current (DC) power, making sure parts are efficient. In a satellite system, you are highly constrained in terms of power while trying to keep the weight down. Consequently, the concept of size, weight, power, and cost (SWaP-C) becomes critical in applications like low-Earth-orbit satellites.

That insight helps drive what we do at Mini-Circuits. Performance is paramount, but the form factor is important too. We are doing more in terms of integrating complex assemblies than we did before. We also do a lot more space and military qualification work. During my time at Amazon, I ran teams doing radiation hardening, so I understood what it means to radiation-harden a satellite. We have a good, growing business at Mini-Circuits in high-reliability and space screening, with lots of opportunity. Those markets need it, and we can provide it cost-effectively. We would like to do more of that in India.

**Across the RF and microwave range, where do you see strategic emphasis in frequency coverage and demand?**

We generally operate from low megahertz, effectively down to DC, up to 110 gigahertz. The density of our product portfolio within any given frequency range is driven primarily by applications and customer demand. Consequently, we maintain an extensive selection of parts in the sub-6 gigahertz spectrum due to high market demand, alongside a robust offering in the Ku-band and selective coverage in the Ka-band moving into millimeter-wave frequencies. We have switches up to 67 gigahertz, and waveguide components up to 110 gigahertz.

> **PERFORMANCE IS PARAMOUNT, BUT THE FORM FACTOR IS IMPORTANT TOO.**

While we maintain capabilities at higher frequencies and the ability to design into millimeter-wave systems, current market demand remains stronger within the lower RF and microwave ranges. However, we are beginning to see increased traction in Frequency Range 3 (FR3), spanning roughly 7 to 15 gigahertz, and are actively developing more parts for that band. Additionally, we are expanding our millimeter-wave footprint around the 40 to 50 gigahertz spectrum.

**What is driving demand in FR3?**

FR3 is emerging as a critical band for 5G-Advanced and potentially 6G deployments, which is significantly driving market demand. Another interesting driver is X-band radar, specifically phased arrays in the 8 to 12 gigahertz range. There is a lot of activity in that band.

**Beyond FR3 and radar, which application segments are spiking for you?**

For us, satellite space applications are really spiking up right now. Some of that is ground radar. In space, we have work in Ku-band or Ka-band, so we are seeing activity there. We also have a strong portfolio of 75-ohm cable television products, and we are seeing a lot of interest in our passive and active devices because of the rollout of the new DOCSIS 4.0 standard. Early FR3 work and radar applications are also driving substantial market demand.

**Are you seeing radar-related demand intersect with automotive electronics?**

We are definitely seeing radar-system activity for automotive. We recently introduced some automotive-qualified parts in our micro-ceramic line, including filters, splitters, couplers, and baluns that are qualified to the AEC-Q200 automotive standard. We already had demand, which is why we did the work, and now we are beginning to see additional traction.

Much of automotive radar technology, such as 77 gigahertz systems, relies on highly integrated circuits. While our components may not be needed within the main chip itself, they are essential for the surrounding RF front-end. We do not manufacture the integrated circuits that power automotive radar for either the 24 gigahertz or 77 gigahertz bands. Our parts are generally complementary to those big chipsets. EFY

This interview is an excerpt of a longer discussion. For the full transcript, please visit: https://efy.co.in/q1tgmv